

Additional Supplemental Material

F Robust Inference

Appendix Table F.1 demonstrates the robustness of our baseline approach to inference in county-level regressions. We consider seven alternative approaches to accounting for spatial autocorrelation in the error terms. Column 1 is our baseline approach of clustering at the level of arbitrary 60 square-mile grid cells following the procedure in [Bester et al. \(2011\)](#). Columns 2–7 adopt the [Conley \(1999\)](#) GMM-based spatial heteroskedasticity-autocorrelation (HAC) consistent procedure that allows for arbitrary correlation in unobservables across all counties within 100, 200, 300, 500, and 1000 km respectively. Column 7 uses two-way clustering on both the arbitrary grid cells in column 1 and the year in which each county entered the frontier. This procedure, based on [Cameron et al. \(2011\)](#), allows for possible correlated unobservable across counties subject to shocks at the same time historically that would have led their county to become relevant to frontier settlers. Finally, column 8 clusters at the state level, using a wild bootstrap procedure to account for the relative small number of states (30 in our main sample). Across all columns, we find no meaningful departures from the precisely estimated effects in our baseline approach. We therefore retain this computationally simple baseline throughout the paper.

Table F.1: Alternative Approaches to Inference

	Base	Spatial HAC: [...] km Bandwidth					Two-Way	State
		100	200	300	500	1000	Base + Yr. Entry	Wild Clust.
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel (a): Infrequent Name Share in 1940								
total frontier experience	0.141 (0.021)	0.141 (0.027)	0.141 (0.027)	0.141 (0.027)	0.141 (0.032)	0.141 (0.040)	0.141 (0.018)	0.141 [0.004]
Number of Counties	2,036	2,036	2,036	2,036	2,036	2,036	2,036	2,036
Panel (b): Infrequent Name Share in 1940, Metaphone-Adjusted								
total frontier experience	0.141 (0.021)	0.141 (0.028)	0.141 (0.029)	0.141 (0.030)	0.141 (0.033)	0.141 (0.041)	0.141 (0.019)	0.141 [0.005]
Number of Counties	2,036	2,036	2,036	2,036	2,036	2,036	2,036	2,036
Panel (c): County Property Tax Rate in 2010								
total frontier experience	-0.034 (0.007)	-0.034 (0.009)	-0.034 (0.009)	-0.034 (0.009)	-0.034 (0.012)	-0.034 (0.009)	-0.034 (0.010)	-0.034 [0.006]
Number of Counties	2,029	2,029	2,029	2,029	2,029	2,029	2,029	2,029
Panel (d): Average Republican Vote Share over 2000–2016								
total frontier experience	2.055 (0.349)	2.055 (0.438)	2.055 (0.469)	2.055 (0.477)	2.055 (0.404)	2.055 (0.249)	2.055 (0.376)	2.055 [0.003]
Number of Counties	2,036	2,036	2,036	2,036	2,036	2,036	2,036	2,036

Notes: This table reports alternative approaches to inference for the four main county-level outcomes used in the paper. The baseline estimates from the paper are reproduced in column 1 for reference. The bracketed numbers in the final column correspond to p-values from a wild bootstrap procedure clustering at the state level.

G Further Background Characterizing Frontier Life

G.1 Demographics and Individualism Along the Transition out of the Frontier

Figure G.1 reports estimates time-varying estimates of equation (1):

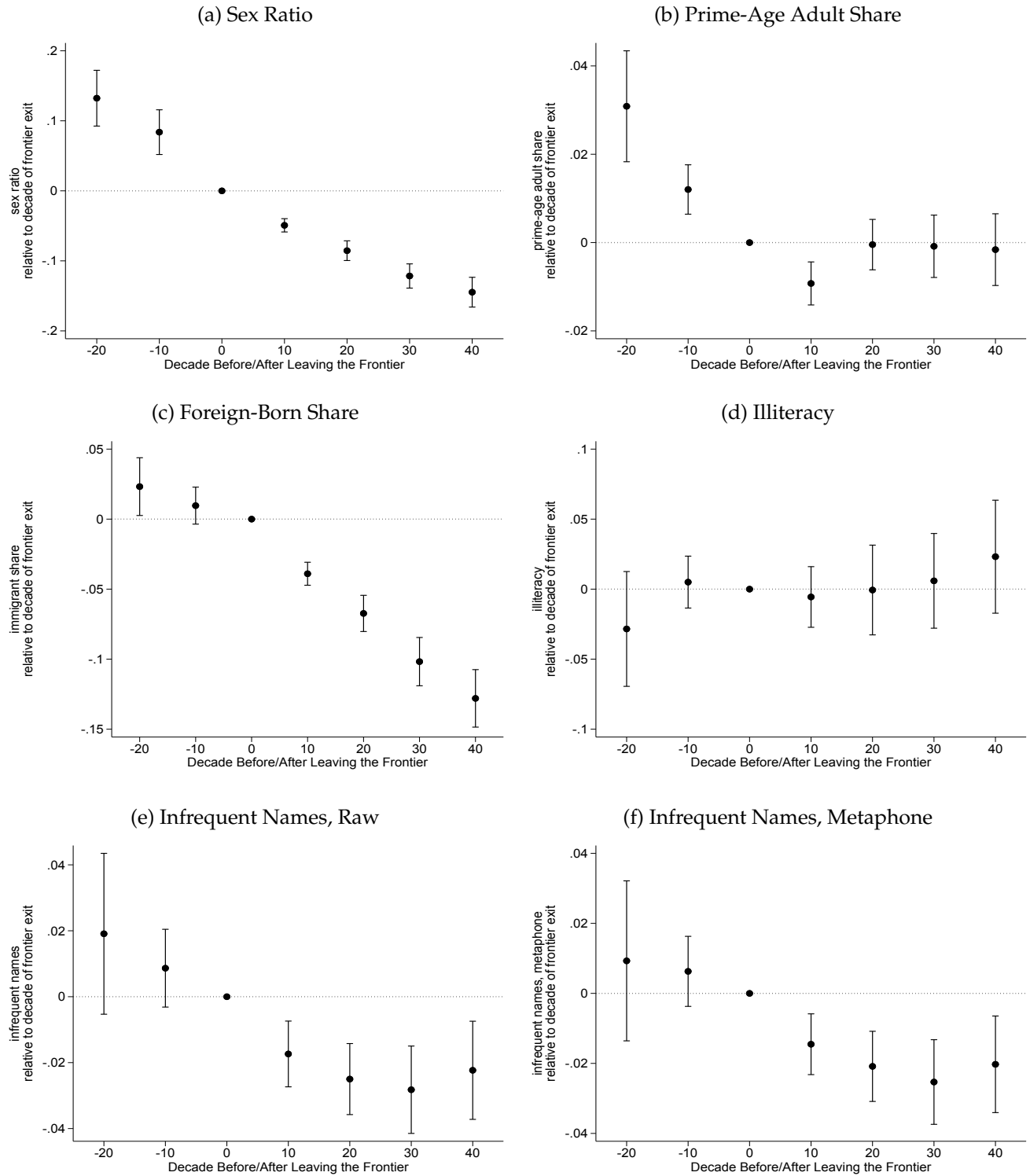
$$x_{cdt} = \alpha + \sum_{j=-20}^{40} \gamma_j \mathbf{1}(\text{years since exiting frontier} = j) + \theta_d + \theta_t + \varepsilon_{cdt}, \quad (\text{G.1})$$

where the γ_j coefficients identify the average x for counties that have exited or will exit the frontier j years prior or in the future, respectively. We plot 95 percent confidence intervals for the γ terms, each of which are estimated with reference to the decade in which the county transitioned out of the frontier.

The estimates in Figure G.1 provide additional insight into the process of demographic and cultural change along the frontier. Panel (a) reveals an abrupt shift in the sex ratio as counties exit the frontier. On average, counties have 0.25 higher sex ratios in the two decades prior to exiting the frontier whereas those decades thereafter exhibit lower ratios that stabilize by the second decade. Panel (b) provides similar evidence of convergence towards a lower prime-age adult share as counties exit the frontier. Panel (c) shows that the foreign-born population share exhibits a steady and roughly linear decline along the frontier transition path. Results are noisier for illiteracy rates in panel (d).

Panels (e) and (f) demonstrate the declining prevalence of individualistic children's names as counties approach the decade in which they exit the frontier. Thereafter, we see naming patterns stabilize around a less individualistic equilibrium in which popular names becomes more common at the local level. Note, however, that these estimates are with reference to the year of exiting the frontier and do not address the long-run differences in individualistic names as a function of years of exposure to frontier conditions. This is precisely what the analysis in Section 4 provides.

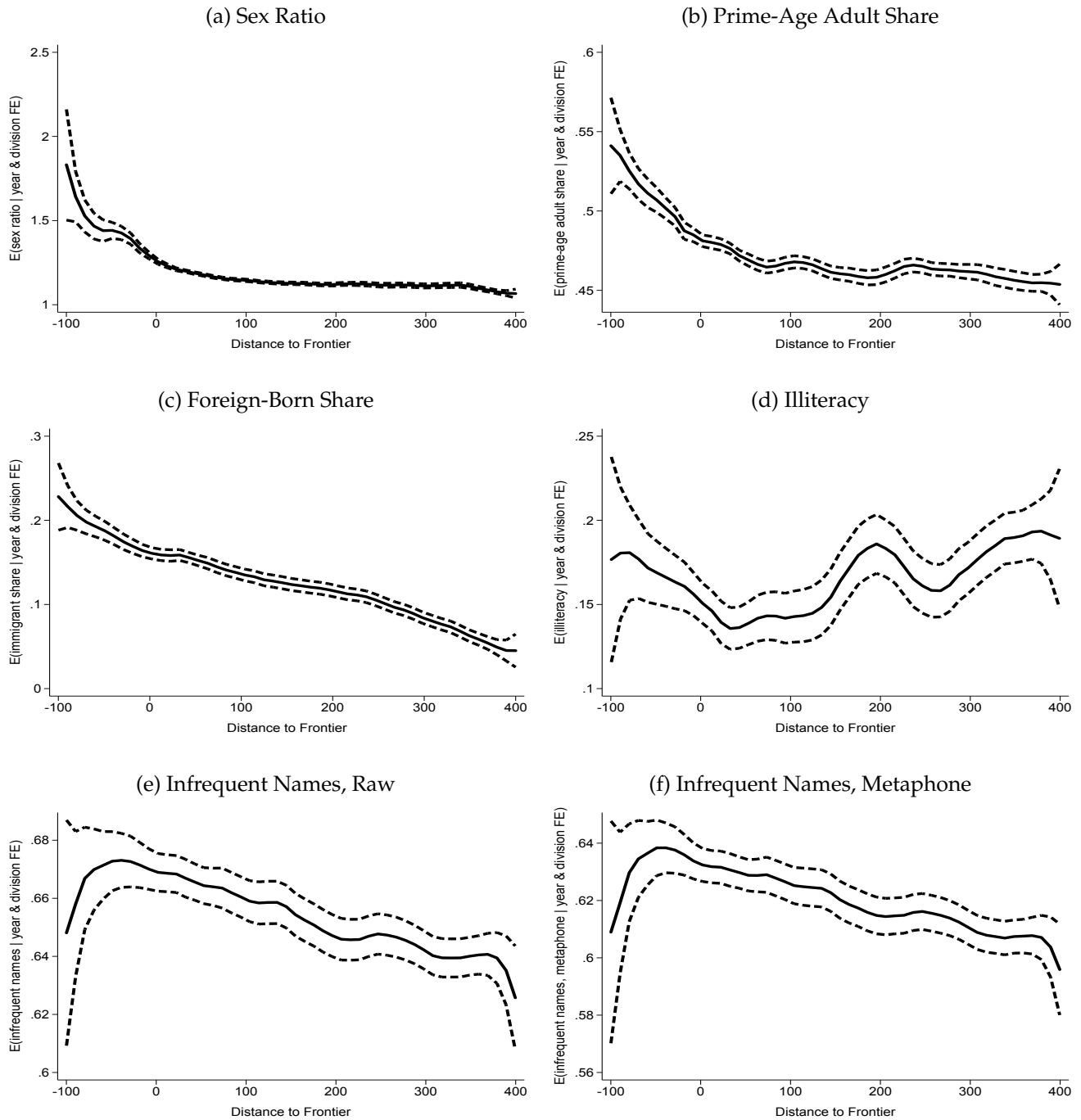
Figure G.1: Demographics and Individualism Along the Transition out of the Frontier
Estimates with Respect to Decade of Exiting the Frontier



Notes: This figure plots coefficients from the event study regressions in equation (G.1) for each of the outcomes in Table 1. The decade-specific point estimates and 95 percent confidence intervals are each with reference to the county-specific decade of exiting the frontier. All regressions include division and year fixed effects. Standard errors are clustered using the grid cell approach of [Bester, Conley and Hansen \(2011\)](#) as described in Section 4.1.

G.2 Demographics and Individualism by Distance to the Frontier

Figure G.2: Distribution of Demographics and Individualism by Distance to the Frontier



Notes: This figure reports the distance-based analogues to the density plots in Figure 4. Distance to the frontier, measured in kilometers, is the distance from the county's centroid to the nearest frontier line. The distance is negative if the county centroid is to the west of the nearest main frontier line. Figures (a)-(f) provide the semiparametric estimates of the corresponding dependent variables, with 95 percent confidence intervals, as a function of distance to the frontier estimated using county-level pooled data and applying a nonlinear function recovered using the partially linear Robinson (1988) estimator. The specification includes Census division and year fixed effects and are based on an Epanechnikov kernel and rule-of-thumb bandwidth.

G.3 Taxes and Public Goods on the Frontier

We present evidence here that demonstrates two important facts about the prevalence of taxation and public goods on the frontier historically.

First, *local* taxation and public spending were significantly lower in frontier counties historically. We demonstrate this for tax revenue per capita in column 1 of Appendix Table G.1 using county- and city-level tax revenue per capita, which is available for 1870 and 1880 as reported by [Manson et al. \(2019\)](#). Similarly, column 2 demonstrates lower government spending per capita in frontier than in non-frontier counties in 1870. Moreover, graphs (a) and (b) in Figure G.3 demonstrate the same sort of structural break in these outcomes at levels of population density around the cutoffs defining frontier locations historically, i.e., 2–6. Together, these results are consistent with smaller local government and weaker taxation on the frontier historically, consistent with the lack of social infrastructure described in most accounts of frontier life.

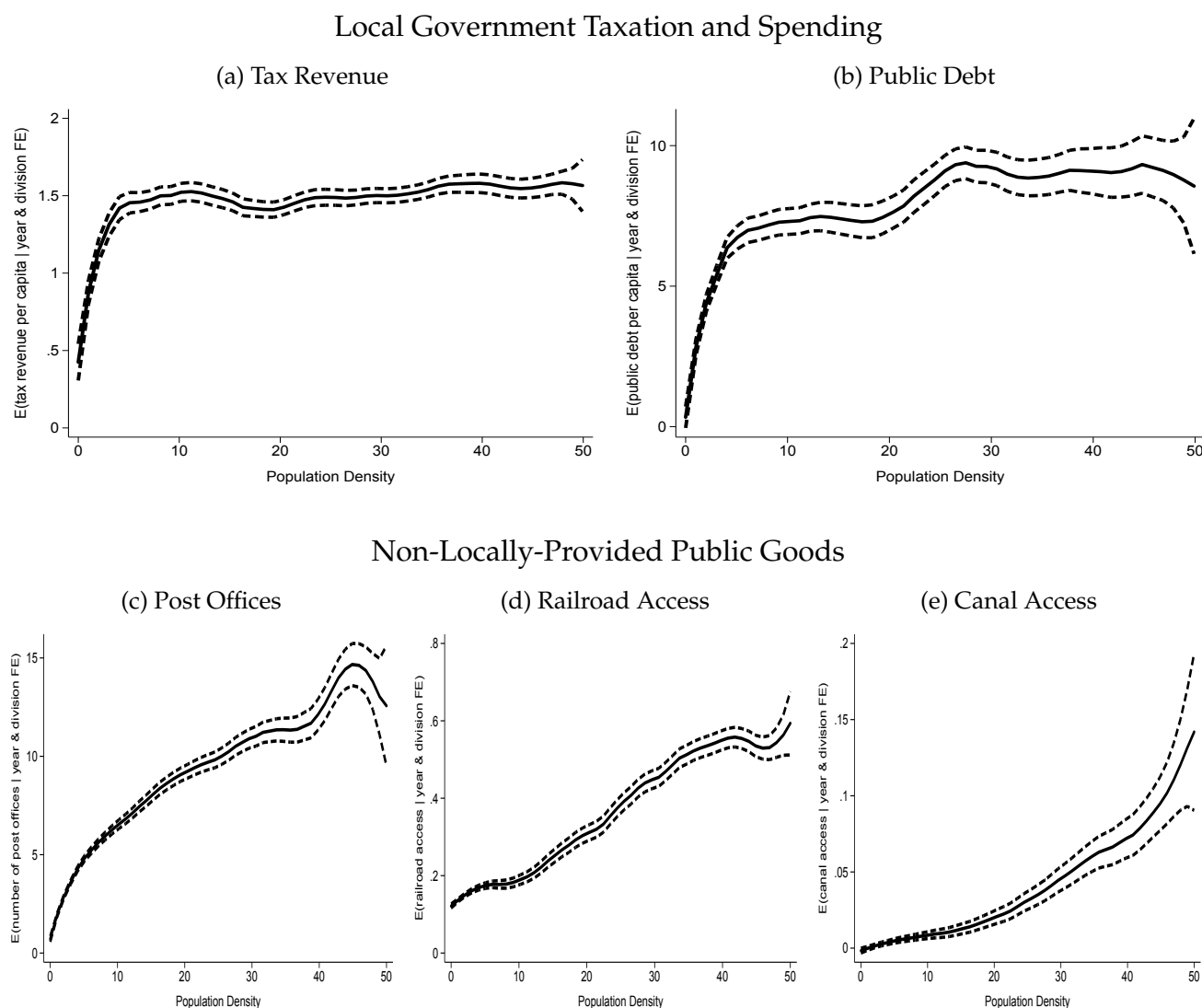
Second, *non-local* public goods were also less pervasive on the frontier, but this infrastructure did not exhibit the same sort of structural break at very low levels of population density seen for local public goods. Columns 3–5 of Appendix Table G.1 show, respectively, that frontier counties had a smaller number of post offices and more limited railroad and canal access historically. Appendix Figure G.3 shows that these non-locally-provided public goods were more pervasive in counties with greater population density. However, unlike local government efforts to raise taxes and make public investments, neither post offices nor railroads and canals exhibit structural changes at low levels of population density characterizing frontier conditions. All three outcomes vary smoothly if not roughly linearly from counties with population density ranging from just above zero to around 40. Importantly, this suggests that non-local government investments were not necessarily leading the westward expansion of the frontier as seen through the types of individuals underlying our results in Sections 3 and 5.

Table G.1: Taxes and Public Goods on the Frontier

Dependent Variable:	Local Government		Non-Local Government		
	Tax Revenue Per Capita (1)	Public Debt Per Capita (2)	Number of Post Offices (3)	Railroad Access (4)	Canal Access (5)
frontier county	-0.565 (0.089)	-4.914 (0.438)	-4.708 (0.360)	-0.114 (0.013)	-0.010 (0.006)
Mean Dep. Var. in Non-Frontier Counties	1.49	8.18	10.02	0.43	0.05
Number of County-Years	3,756	1,789	9,801	13,458	14,043
R ²	0.42	0.16	0.44	0.54	0.15

Notes: This table reports estimates of the historical frontier differential in public goods using the same specification as in Table 1. Column 1 is the inverse hyperbolic sine of the sum of `taxcountyppop` and `taxcityetcppop` reported by [Manson et al. \(2019\)](#) for 1870 and 1880. Column 2 is the inverse hyperbolic sine of the sum of `pubdebtcounty` and `pubdebtcityetc`. Column 3 is the number of post offices in the given county–decade as reported in [Acemoglu et al. \(2016\)](#). Columns 4 and 5 are indicators equal to one if the given county–decade has railroad and canal access, respectively. See the notes below that table for details on the specification.

Figure G.3: Taxes and Public Goods by Population Density, 1790–1890



Notes: This figures reports semiparametric estimates of the relationship between population density and public goods measures from Table G.1 using the same specification as in Figure 4. See the notes below that table for details on the specification.

G.4 Low Inequality and Effort as the Road to Riches on the Frontier

This section argues that the opportunities and challenges on the frontier contributed to a culture of opposition to government intervention. The frontier's favorable prospects of upward mobility and a large perceived importance of effort in income generation may have fostered opposition to taxes, as suggested by the political economy theories of preferences for redistribution. This connection between the American frontier and theories of preferences for redistribution, hinted at by [Alesina et al. \(2001\)](#), echoes [Billington \(1974\)](#), who argued that the frontiersman "wanted not government interference with his freedom as he followed the road to riches."

In his analysis of the Turner thesis, [Billington \(1974\)](#) emphasizes the implications of the frontier's land abundance and "widespread property holdings." In these conditions, "a man's capacities, not his ancestry, determined his eventual place in the hierarchy, to a greater degree than in older societies." The frontiersman believed that "his own abilities would assure him a prosperous future as he exploited the natural resources about him." Access to land offered profit of opportunities, even for settlers with low

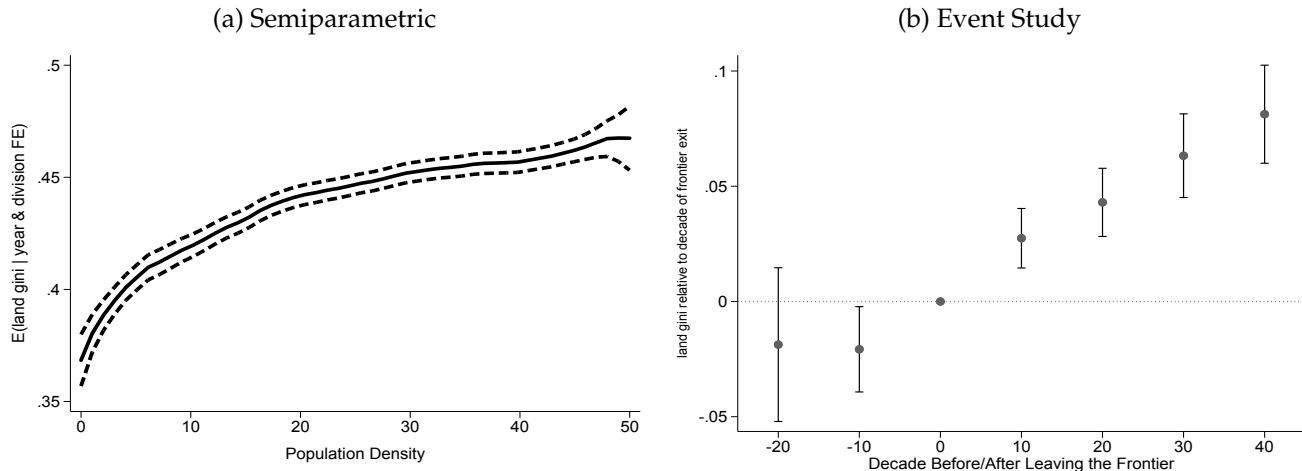
initial wealth. Class distinctions were also weakened by the ubiquity of threats characterizing frontier life. As [Overmeyer \(1944\)](#) argues, since everyone “had to face the same hardships and dangers,” the frontier was a “great leveling institution.”

Numerous historical studies present stylized facts consistent with favorable prospects of upward mobility on the frontier and also with a large perceived importance of effort. As summarized by [Stewart \(2006\)](#), the frontier was “a place of economic opportunity,” where settlers had low levels of initial wealth, but land-holding was widespread and rates of wealth accumulation were high, especially for early settlers and those that were able to endure.

Indeed, as shown in Appendix Figure [G.4](#), historical Census data on landholdings is consistent with the idea that frontier locations offered a more level playing field. Land inequality, captured by the Gini coefficient, was significantly lower on the frontier, with sharp differences resembling what we documented for key demographics and individualism in Section 3. Moreover, this difference dissipated over time as counties exited the frontier and the usual forces giving rise to inequality took hold.

In sum, the stylized facts summarized above suggest a relatively limited role for inherited social class as a key determinant of income and wealth generation in the frontier economy. This implied a level playing field offering equality of opportunity, and a relatively high importance of effort as opposed to luck (of being born into a given class). Together with the selection and cultivation of individualism, these conditions plausibly contributed to the origins and persistence of frontier culture.

Figure G.4: Inequality is Lower on the Frontier

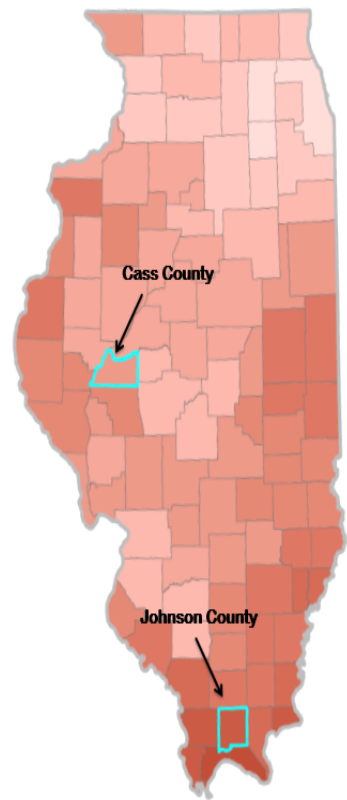


Notes: Based on county level data from [Manson et al. \(2019\)](#) from 1790-1890. Land inequality is measured using the county level gini coefficient based on the number of farms in seven bins of farm size. The semiparametric specification in (a) is the same as in Figure 4, and the event study specification in (b) is the same as in Figure [G.1](#). See the notes therein for details.

H Case Study Illustrating Long-Run Effects

To fix ideas, consider the two counties of Cass and Johnson mentioned in Section 2.2 and seen in the TFE map on the right, which is a snapshot of Illinois from Figure 3. Both are roughly equidistant from the Mississippi River and the important historical city of St. Louis. Today, the two rural counties look very similar in terms of population density: Cass has 36.3 people/mi², and Johnson has 36.6 people/mi². These two counties had very similar population density in 1890 as well. However, they differ significantly in their total frontier experience historically. Cass was on the frontier for 10 years, and Johnson for 32 years. This difference may be explained by any number of factors shaping the westward movement of the frontier through this area of the midwest in the early 1800s as seen in Figure 2. One potentially important contributor lies in our instrumental variable. Johnson entered the frontier in 1803 whereas Cass entered in 1818. While only 15 years apart, this implied a considerable difference in exposure to subsequent immigration-induced pressure on the westward expansion of the frontier over the next few decades as evidenced in Figures D.1, D.2, and especially D.3.

These historical differences in TFE translate into substantial long-run differences in the prevalence of rugged individualism in local culture. In Cass, 75 (64) percent of girls (boys) have infrequent names in 1940, Republican presidential candidates captured 55 percent of the vote in the average election since 2000, and local property tax rates are around 1.9 percent in 2010. Meanwhile, in Johnson, 78 (71) percent of girls (boys) have infrequent names in 1940, 68 percent average Republican vote shares since 2000, and 1.3 percent local property tax rates in 2010. This is striking insomuch as the two counties have such similar contemporary population density.



I Other Evidence on Selective Migration

This section provides additional evidence on selective migration and returns to individualism using a linked sample of men from 1870 to 1880. These results are discussed at greater length in an earlier working paper, but we retain them here given that they complement the more comprehensive results in Section 5 of the paper.

Compared to the results in the paper, those presented below are based on a narrower, linked sample of households from 1870 to 1880. We create this sample using complete-count restricted-access data from IPUMS (Ruggles et al., 2019) for 1870 and the US 100% sample (Ruggles et al., 2019) from the North Atlantic Population Project (NAPP) from Minnesota Population Center (2019) for 1880. We focus on the latest consecutive rounds available within the frontier period to ensure a large sample. We link individuals across rounds using an algorithm developed by Feigenbaum (2016). The base sample in 1880 is restricted to male household heads, native-born, aged 30–50, white, and who have at least one child aged 0–10.¹

Appendix Table I.1 reports estimates of the frontier differential in infrequent naming patterns based on versions of the following equation for different sub-populations of movers and stayers:

$$\text{child has infrequent name}_{ic,1880} = \alpha + \beta \text{frontier}_{c,1880} + \mathbf{x}'_{ic}\boldsymbol{\zeta} + \varepsilon_{ic,1880}, \quad (\text{I.1})$$

where the binary dependent variable equals one if child i residing in county c in 1880 has a name that falls outside the top 10 nationally in that decade, and the frontier indicator equals one if county c lies on the frontier according to our baseline definition. We restrict attention to white children aged 0–10 with native-born parents and cluster standard errors at the county level. The $\mathbf{x}_{ic}$ vector includes age $\times$ gender and birth order fixed effects as well as indicators for whether the parents have infrequent names, but results are identical without these controls.

Column 1 of Appendix Table I.1 identifies the significance of selective migration. Children in households that migrated to the frontier between 1870 and 1880 are 4.2 p.p. more likely to have infrequent names than those remaining in non-frontier areas during that period, 71 percent of whom have infrequent names. While we do not observe whether these children were born before or after arriving on the frontier, this differential points to the self-selection of individualist types.

Column 2 captures the overall frontier differential in individualism. Children in frontier counties in 1880 are 7.5 p.p. more likely to have an infrequent name relative to children in non-frontier locations. Next, we show that the longer-term frontier residents (stayers) exhibit stronger individualism than recent arrivals from other counties. Column 3 decomposes the 7.5 p.p. differential into differences coming from early versus later frontier settlers. Early settlers in frontier counties are nearly three times more likely to give their children infrequent names than those that arrived more recently during the 1870s. Column 4 corroborates this differential, restricting the sample to those living in frontier counties in 1880. These results suggest that greater time on the frontier is associated with more individualistic naming patterns, a result that is rigorously borne out in Section 5.2 of the paper.

¹The target year is 1870. The set of potential matches for these men are first identified based on first and last name, birth state and birth year. A random training sample is then drawn from among the potential matches and manually trained. The importance of each match feature is quantified using a probit model, and used to estimate a probability score for each link. A true match is defined as one with a sufficiently high score both in absolute and relative terms. The match rate was 25 percent, which is comparable with the rates achieved by recent studies linking records with broadly comparable data albeit different target populations (e.g., 29 percent in Abramitzky et al., 2012; 26 percent in Collins and Wanamaker, 2017; and 22 percent in Long and Ferrie, 2013). Although matching on names leaves scope for sample selection, the results look similar when reweighting using the odds of being linked across Census rounds (following Bailey et al., forthcoming). We estimate these probabilities using the same characteristics used for linking as well as an interaction of infrequent name status and frontier location in 1880. These interactions re-balance the linked sample to account for differential missing-ness along our key variables of interest.

Table I.1: Frontier Individualism and Selective Migration

	Dependent Variable: Child Has Infrequent Name in 1880			
	(1)	(2)	(3)	(4)
omitted reference group:	non-frontier resident, 1870–80	non-frontier resident, 1880	non-frontier resident, 1880	frontier immigrant, 1870–80
frontier county resident in 1880, immigrant	0.042 (0.012)		0.055 (0.011)	
frontier county resident in 1880		0.075 (0.018)		
frontier county resident in 1880, non-immigrant			0.186 (0.035)	0.118 (0.026)
Number of Individuals	1,223,600	1,239,513	1,239,513	12,630
Mean Infrequent Name Share	0.707	0.708	0.708	0.767
R ²	0.02	0.02	0.02	0.05
Gender×Age Fixed Effects	✓	✓	✓	✓
Birth Order Fixed Effects	✓	✓	✓	✓

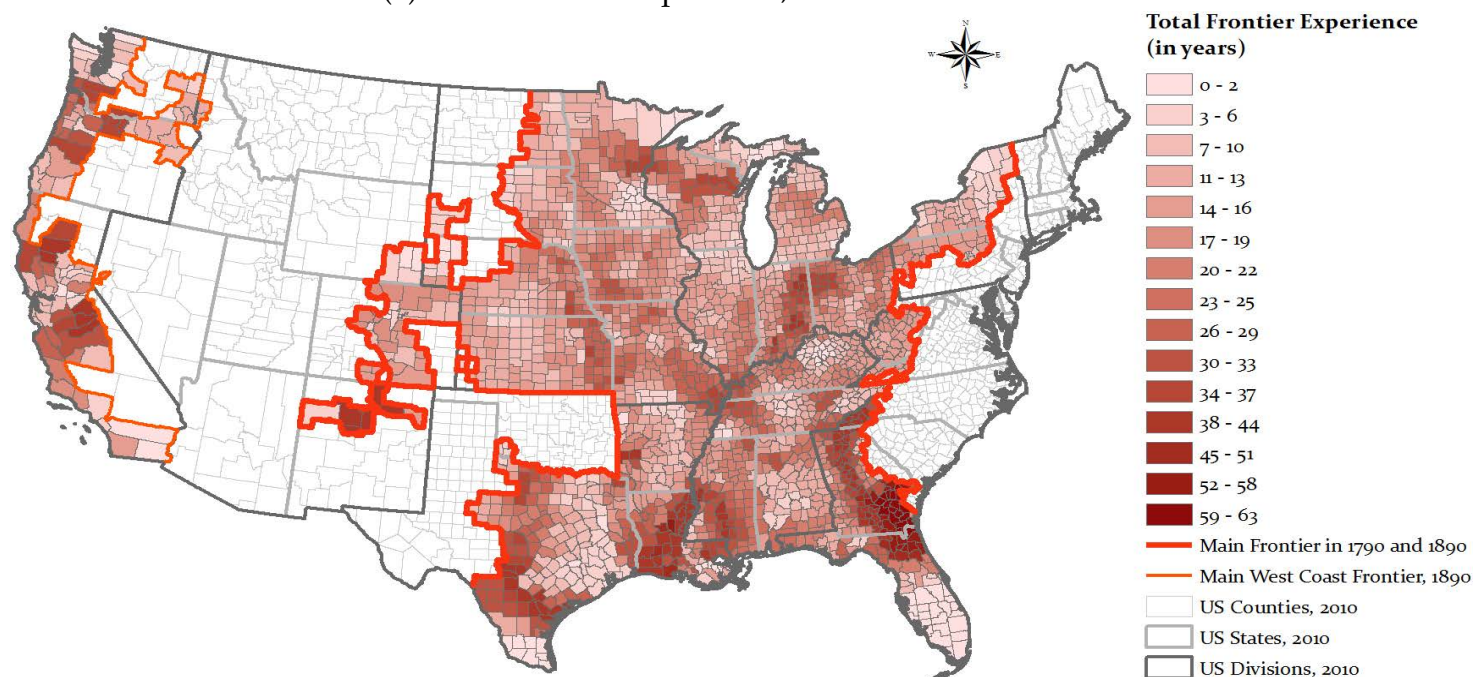
Notes: This table reports estimates of equation (I.1) based on the linked historical Census data from 1870 to 1880 for households with white, native-born fathers age 30–50 and children aged 0–10 in 1880. The dependent variable is an indicator equal to one if the child is given a name that falls outside the top 10 most popular names nationally in the 1870s. The top of each column reports the omitted reference group and the mean infrequent name share among them. We define immigrant status here based on whether the father switched counties between 1870 and 1880. Frontier counties are as defined in 1870 and 1880 based on the main east-to-west frontier line. Column 1 reports the selective migration differential between migrants from non-frontier to frontier counties and those that remained in non-frontier counties in both 1870 and 1880. Column 2 reports the overall differential in infrequent names between frontier and non-frontier counties in 1880, i.e., inclusive of stayers in frontier counties. Column 3 breaks down the overall differential into the component due to migrants between 1870 and 1880 and those that resided in the frontier county prior to 1870 (either by birth or earlier migration). Column 4 then restricts to frontier county residents, identifying the differential between recent immigrants and longer-term residents. In addition to gender×age and birth order fixed effects, all regressions control for indicators for whether the mother and father have infrequent names. Standard errors are clustered by county in 1870.

J Additional Empirics

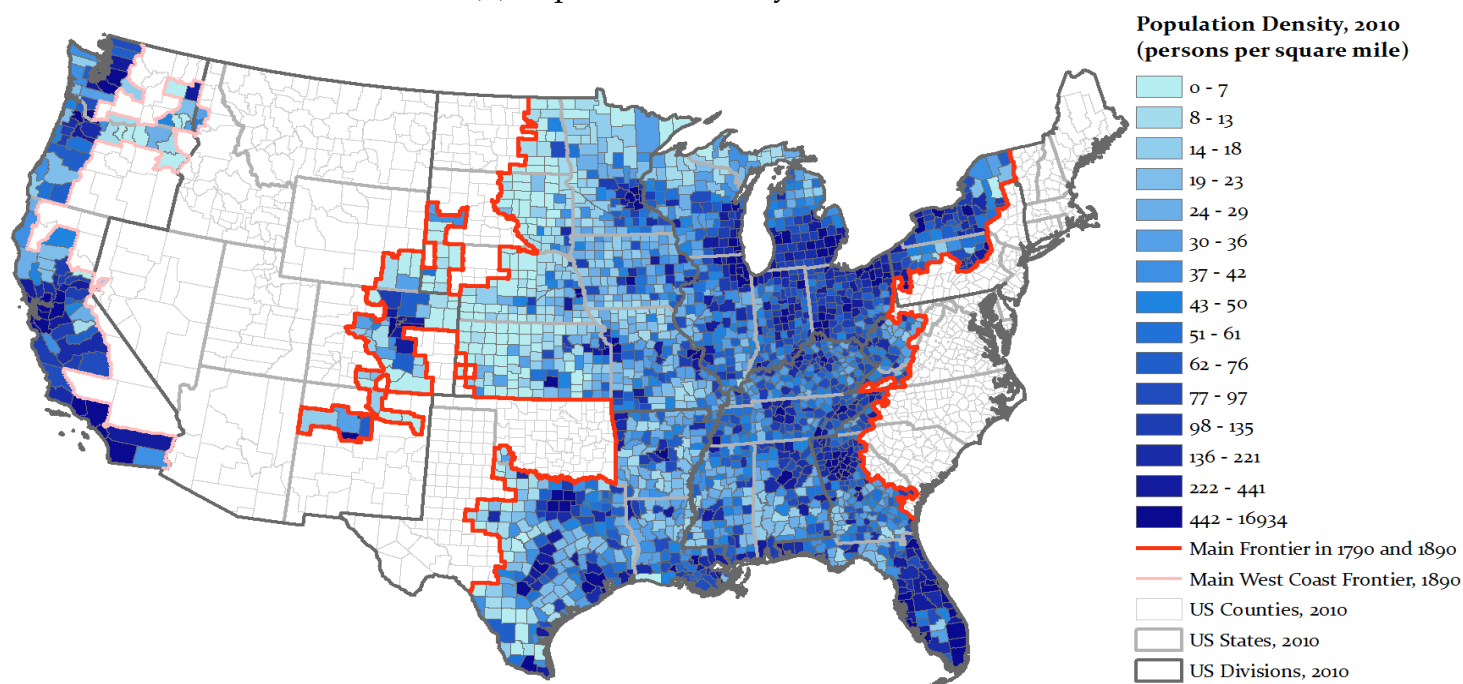
J.1 Comparing Total Frontier Experience and Current Population Density

Figure J.1: TFE is Distinct from Current Population Density

(a) Total Frontier Experience, 1790 to 1890



(b) Population Density, 2010



Notes: Panel (a) reproduces Figure 3, and (b) presents a similarly scaled map of population density in 2010 for the same counties.

J.2 Alternative Measures of Total Frontier Experience

Appendix Table J.1 reports baseline results excluding the 40 counties with zero TFE in the data. As noted in the paper, the findings are not sensitive to excluding this extensive margin of variation in TFE.

Our baseline measure of TFE closely followed definitions in the historical literature as discussed in Section 2. In Table J.2, we demonstrate the robustness of our results to three relevant margins of adjustment to our measure of TFE. In each case, we redefine what it means for county c to be on the frontier at time t . First, we reduce the catchment area from 100 km to 50 km in proximity to the frontier line. Second, we adjust the density restriction to include counties with > 2 people/mi² but still less than 6, counties with ≤ 18 people/mi², and then remove the population density restriction altogether. Finally, we consider defining the frontier line as including only the main, westernmost extent of all contour lines identified by the GIS algorithm. The overall message is that our particular choice of the frontier definition based on the historical record is not driving the main findings.

Table J.1: Robustness to Excluding Counties with Zero TFE

Dependent Variable:	Infrequent Name raw metaphone standardized (1)	(2)	Mean Gov. Prefs. CCES (3)	County Property Tax Rate (4)	Republican Vote Share Avg. (5)
Panel (a): Baseline					
total frontier experience	0.141 (0.021)	0.141 (0.021)	0.014 (0.002)	-0.034 (0.007)	2.055 (0.349)
Number of Observations	2,036	2,036	112,759	2,029	2,036
Mean of Dependent Variable	0.00	0.00	0.41	1.02	60.04
Panel (b): Excluding Counties with Zero TFE					
total frontier experience	0.137 (0.020)	0.135 (0.021)	0.013 (0.002)	-0.031 (0.007)	1.664 (0.340)
Number of Observations	1,996	1,996	105,623	1,990	1,996
Mean of Dependent Variable	0.00	0.00	0.42	1.02	60.30

Notes: This table reports estimates in panel (b) of our baseline specification (reproduced in panel (a)) excluding up to 40 counties with zero TFE. The specification is otherwise

Table J.2: Robustness to Alternative Measures of TFE for Summary Outcomes

Dependent Variable:	Infrequent Name raw standardized 1940 (1)	metaphone 1940 (2)	Mean Gov. Prefs. CCES 2006–16 (3)	County Property Tax Rate 2010 (4)	Republican Vote Share Avg. 2000–16 (5)
TFE: 100 km, $\leq 6/\text{mi}^2$, no inner or outer islands	0.141 (0.021)	0.141 (0.021)	0.014 (0.002)	-0.034 (0.007)	2.055 (0.349)
TFE: 50 km, $\leq 6/\text{mi}^2$, no inner or outer islands	0.144 (0.021)	0.147 (0.022)	0.017 (0.003)	-0.035 (0.007)	2.051 (0.358)
TFE: 100 km, $\leq 18/\text{mi}^2$, no inner island lines	0.106 (0.021)	0.105 (0.022)	0.010 (0.002)	-0.027 (0.007)	1.575 (0.339)
TFE: 50 km, $\leq 18/\text{mi}^2$, no inner island lines	0.093 (0.020)	0.100 (0.020)	0.011 (0.003)	-0.025 (0.006)	1.458 (0.351)
TFE: 100 km, 2-6/ mi^2 , no inner island lines	0.099 (0.035)	0.109 (0.035)	0.025 (0.006)	-0.014 (0.008)	1.877 (0.485)
TFE: 50 km, 2-6/ mi^2 , no inner island lines	0.080 (0.039)	0.086 (0.039)	0.026 (0.006)	-0.012 (0.009)	1.771 (0.530)
TFE: 100 km, no density restriction, no inner island lines	0.034 (0.021)	0.026 (0.021)	0.006 (0.002)	-0.011 (0.007)	1.001 (0.335)
TFE: 50 km, no density restriction, no inner island lines	0.057 (0.019)	0.062 (0.019)	0.006 (0.003)	-0.018 (0.006)	1.078 (0.339)
TFE: 100 km, $\leq 6/\text{mi}^2$, including inner island lines	0.162 (0.019)	0.162 (0.020)	0.014 (0.002)	-0.032 (0.006)	2.048 (0.320)
TFE: 50 km, $\leq 6/\text{mi}^2$, including inner island lines	0.172 (0.021)	0.178 (0.023)	0.017 (0.003)	-0.035 (0.007)	2.098 (0.335)
TFE: 100 km, $\leq 6/\text{mi}^2$, main single contour line	0.112 (0.027)	0.106 (0.028)	0.011 (0.004)	-0.037 (0.008)	1.872 (0.436)
TFE: 50 km, $\leq 6/\text{mi}^2$, main single contour line	0.106 (0.029)	0.106 (0.029)	0.018 (0.004)	-0.043 (0.008)	1.787 (0.460)
TFE: 50 km, $\leq 6/\text{mi}^2$, no inner or outer island lines	0.139 (0.021)	0.136 (0.022)	0.012 (0.002)	-0.034 (0.007)	2.133 (0.357)
TFE: 50 km, $\leq 6/\text{mi}^2$, no inner or outer island lines	0.139 (0.022)	0.136 (0.023)	0.016 (0.003)	-0.035 (0.007)	2.116 (0.373)
State Fixed Effects	✓	✓	✓	✓	✓
Geographic/Agroclimatic Controls	✓	✓	✓	✓	✓

Notes: This table reports estimates of equation (4) for three measures of infrequent names for white children, age 0–10 in the 1940 Census. Each cell is a different regression based on the given dependent variable in the column and the measure of total frontier experience in the given row. The frontier lines considered in the baseline are countour lines longer than 500km after removing all “inner island lines” that are east of the main frontier line. The alternative measures of frontier experience considered above vary (i) the catchment area from 100 to 50 km from the contour lines, (ii) the density restriction from $\leq 6 \text{ people}/\text{mi}^2$ to $2 \geq \text{people}/\text{mi}^2 \leq 6$ to no restriction, (iii) including inner island lines, and (iv) including only the longest single contour line. Standard errors are clustered based on the grid-cell approach of [Bester, Conley and Hansen \(2011\)](#) as detailed in Section 4.1.

J.3 Full Elaboration of Additional Controls in Table B.6

Table J.3: Robustness to Additional Controls

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
Panel (a): Infrequent Name Share in 1940 (standardized)														
total frontier experience	0.141 (0.021)	0.143 (0.020)	0.139 (0.021)	0.143 (0.020)	0.141 (0.021)	0.145 (0.021)	0.106 (0.021)	0.142 (0.020)	0.134 (0.019)	0.107 (0.020)	0.127 (0.022)	0.094 (0.022)	0.151 (0.020)	0.089 (0.019)
ruggedness		4.965 (8.873)												3.411 (6.602)
rainfall risk			2.515 (1.800)											17.428 (1.438)
distance to nearest portage site				0.047 (0.045)										0.038 (0.032)
distance to nearest mineral discovery site					0.093 (0.026)									0.098 (0.022)
distance to nearest Indian battle						-0.043 (0.037)								-0.060 (0.025)
slave population share, 1860							-0.013 (0.002)							-0.008 (0.001)
sex ratio, 1890								-0.391 (0.130)						-0.258 (0.080)
immigrant share, 1890									-0.008 (0.005)					0.033 (0.009)
scottish/irish immigrant share, 1890										-0.274 (0.021)				-0.192 (0.020)
birthplace diversity, 1890											-1.256 (0.272)			-1.774 (0.611)
years connected to railroad by 1890												-0.021 (0.002)		-0.012 (0.002)
manufacturing employment share, 1890													-0.079 (0.006)	-0.056 (0.005)
Order β for $\beta = 0$	-13.80	-11.03	-40.85	-8.28	-14.43	-6.38	1.23	-13.21	10.07	2.03	3.86	1.16	-6.57	1.61
Number of Counties	2,036	2,036	2,036	2,036	2,036	2,036	2,036	2,036	2,036	2,036	2,036	2,036	2,036	2,036
R ²	0.60	0.62	0.60	0.60	0.61	0.60	0.62	0.61	0.61	0.67	0.61	0.66	0.66	0.76
Panel (b): Infrequent Name Share in 1940: Metaphone-Adjusted (standardized)														
total frontier experience	0.141 (0.021)	0.143 (0.021)	0.139 (0.021)	0.143 (0.021)	0.141 (0.021)	0.144 (0.021)	0.107 (0.022)	0.141 (0.021)	0.135 (0.020)	0.106 (0.020)	0.128 (0.021)	0.094 (0.022)	0.150 (0.021)	0.091 (0.019)
ruggedness		5.282 (9.910)												3.865 (6.442)
rainfall risk			2.150 (1.914)											-0.820 (1.530)
distance to nearest portage site				0.065 (0.046)										0.058 (0.036)
distance to nearest mineral discovery site					0.105 (0.028)									0.109 (0.024)
distance to nearest Indian battle						-0.035 (0.038)								-0.052 (0.028)
slave population share, 1860							-0.012 (0.002)							-0.008 (0.001)
sex ratio, 1890								-0.394 (0.141)						-0.264 (0.086)
immigrant share, 1890									-0.007 (0.005)					0.025 (0.010)
scottish/irish immigrant share, 1890										-0.281 (0.021)				-0.213 (0.021)
birthplace diversity, 1890											-1.109 (0.279)			-1.016 (0.720)
years connected to railroad by 1890												-0.021 (0.002)		-0.012 (0.002)
manufacturing employment share, 1890													-0.078 (0.007)	-0.055 (0.006)
Order β for $\beta = 0$	-16.20	-12.62	-36.47	-8.30	-17.00	-8.30	1.49	-13.37	16.69	2.21	3.20	1.30	-7.25	1.93
Number of Counties	2,036	2,036	2,036	2,036	2,036	2,036	2,036	2,036	2,036	2,036	2,036	2,036	2,036	2,036
R ²	0.58	0.60	0.58	0.58	0.58	0.58	0.59	0.58	0.58	0.65	0.59	0.63	0.63	0.73
Panel (c): Mean Government Preferences Outcomes (CCES), 2006–16														
total frontier experience	0.014 (0.002)	0.013 (0.002)	0.014 (0.002)	0.014 (0.002)	0.014 (0.002)	0.013 (0.002)	0.014 (0.002)	0.014 (0.002)	0.011 (0.002)	0.011 (0.002)	0.010 (0.002)	0.012 (0.002)	0.011 (0.002)	0.009 (0.003)
ruggedness		-0.342 (0.111)												-0.306 (0.129)
rainfall risk			-0.371 (0.296)											-0.203 (0.320)
distance to nearest portage site				0.000 (0.000)										-0.000 (0.000)
distance to nearest mineral discovery site					0.009 (0.004)									0.007 (0.004)
distance to nearest Indian battle						0.005 (0.005)								0.002 (0.005)
slave population share, 1860							0.000 (0.000)							0.000 (0.000)
sex ratio, 1890								0.008 (0.010)						-0.003 (0.019)
immigrant share, 1890									-0.002 (0.000)					0.001 (0.002)
scottish/irish immigrant share, 1890										-0.013 (0.002)				-0.005 (0.003)
birthplace diversity, 1890											-0.150 (0.025)			-0.094 (0.118)
years connected to railroad by 1890												-0.001 (0.000)		-0.001 (0.000)
manufacturing employment share, 1890													-0.004 (0.001)	-0.002 (0.001)
Order β for $\beta = 0$	6.49	6.23	7.07	7.00	6.34	5.46	7.00	6.66	3.73	3.36	3.33	4.20	3.71	2.26
Number of Counties	112,759	112,759	112,759	112,759	112,759	112,759	112,759	112,759	112,759	112,759	112,759	112,759	112,759	112,759
Mean of Dependent Variable	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41
R ²	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05
Panel (d): County Property Tax Rate in 2010														
total frontier experience	-0.034 (0.007)	-0.034 (0.006)	-0.034 (0.007)	-0.033 (0.006)	-0.034 (0.006)	-0.033 (0.006)	-0.033 (0.007)	-0.034 (0.007)	-0.028 (0.006)	-0.026 (0.006)	-0.027 (0.006)	-0.023 (0.006)	-0.036 (0.006)	-0.023 (0.006)
ruggedness		-0.959 (0.219)												-0.724 (0.230)
rainfall risk			0.959 (0.581)											0.736 (0.516)
distance to nearest portage site				0.026 (0.019)										0.006 (0.016)
distance to nearest mineral discovery site					0.026 (0.009)									0.024 (0.008)
distance to nearest Indian battle						-0.004 (0.011)								0.002 (0.009)
slave population share, 1860							0.000 (0.000)							-0.001 (0.003)
sex ratio, 1890								-0.010 (0.027)						-0.024 (0.024)
immigrant share, 1890									0.007 (0.001)					-0.003 (0.003)
scottish/irish immigrant share, 1890										0.059 (0.008)				0.029 (0.006)
birthplace diversity, 1890											0.636 (0.084)			0.341 (0.244)
years connected to railroad by 1890												0.005 (0.000)		0.003 (0.000)
manufacturing employment share, 1890													0.016 (0.002)	0.008 (0.002)
Order β for $\beta = 0$	-27.45	-7.11	-3.23	2.54	-30.95	9.19	8.39	-30.17	0.35	0.48	0.49	0.28	-2.08	0.42
Number of Counties	2,036	2,036	2,036	2,036	2,036	2,036	2,036	2,036	2,036	2,036	2,036	2,036	2,036	2,036
Mean of Dependent Variable	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02	1.02
R ²	0.82	0.82	0.82	0.82	0.82	0.82	0.82	0.82	0.83	0.83	0.83	0.83	0.83	0.85
Panel (e): Republican Vote Share, Average 2000–16														
total frontier experience	2.055 (0.349)	2.050 (0.349)	2.115 (0.338)	2.095 (0.344)	2.055 (0.350)	2.172 (0.351)	1.399 (0.361)	2.060 (0.347)	1.715 (0.328)	1.717 (0.340)	1.689 (0.327)	1.640 (0.361)	2.137 (0.350)	0.931 (0.256)
ruggedness			-11.710 (11.247)											-34.668 (9.508)
rainfall risk				-77.882 (42.010)										-108.600 (36.779)
distance to nearest portage site					0.930 (0.777)									0.566 (0.623)
distance to nearest mineral discovery site						0.191 (0.406)								0.137 (0.344)
distance to nearest Indian battle							-1.393 (0.603)							-0.794 (0.471)
slave population share, 1860								-0.237 (0.035)						-0.270 (0.034)
sex ratio, 1890									-5.000 (1.856)					-1.948 (0.940)
immigrant share, 1890										-0.417 (0.051)				-0.217 (0.198)
scottish/irish immigrant share, 1890											-2.738 (0.263)			-0.243 (0.243)
birthplace diversity, 1890												-32.982 (3.563)		-4.970 (12.266)
years connected to railroad by 1890													-0.186 (0.026)	-0.089 (0.025)
manufacturing employment share, 1890													-0.689 (0.090)	-0.422 (0.075)
Order β for $\beta = 0$	-8.55	-8.93	-6.43	-6.75	-8.36	-5.10	2.34	-8.50	7.33	7.40	6.72	4.53	-7.81	1.35
Number of Counties	60,04	60,04	60,04	60,04	60,04	60,04	60,04	60,04	60,04	60,04	60,04	60,04	60,04	60,04
Mean of Dependent Variable	0.33	0.33	0.34	0.33	0.33	0.34	0.38	0.34	0.38	0.38	0.39	0.36	0.36	0.49
R ²	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
State Fixed Effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Geographic/Agricultural Controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Notes: This table reproduces the estimates from Table B.6, showing the coefficient estimates for the additional variables listed at the top of that table. Standard errors are clustered based on the grid-cell approach of Bester, Conley and Hansen (2011) as detailed in Section 4.1.

J.4 The Parasite-Stress Theory of Values

The parasite-stress theory of values due to [Thornhill and Fincher \(2014\)](#) argues that the prevalence of infectious diseases leads to higher levels of in-group assortative sociality, which they associate with collectivism, as an adaptive response that minimizes contagion. In the context of our study, this theory might suggest that frontier individualism resulted from the low prevalence of infectious diseases on the frontier. However, this potential mechanism does not arise in historical narratives. Nor do we find evidence of differential disease prevalence or morbidity on the frontier. As seen in Table J.4 below, the prevalence of pathogens—associated with tuberculosis, malaria, and typhoid, among other diseases considered in [Gorodnichenko and Roland \(2016\)](#)—does not exhibit any differential intensity on the frontier. We can measure the incidence of these specific infectious diseases as well as a broad array of other illnesses for the first time in the 1880 Population Census. Adopting specifications similar to Table 1, we find little evidence that individuals living on the frontier had differential (infectious) disease or illness. If the parasite-stress mechanism were salient, we would find that frontier locations exhibit significantly less prevalence of infectious diseases. While the relatively precise zeros in the table may be specific to 1880, this provides suggestive evidence that the parasite-stress channel is not a first-order factor in explaining the differential individualism on the frontier.

Table J.4: No Differential Infectious Diseases or Sickness on the Frontier

Dependent Variable:	Share of Pop. with Infectious Disease		Share of Pop. with Any Illness	
	(1)	(2)	(3)	(4)
on the frontier	0.0001 (0.0003)		0.0009 (0.0013)	
near frontier line		-0.0001 (0.0001)		0.0011 (0.0007)
low population density		0.0000 (0.0002)		0.0001 (0.0010)
Mean Dependent Variable	0.001	0.001	0.009	0.009
Number of Counties	1,780	1,780	1,780	1,780
R ²	0.05	0.05	0.08	0.08

Notes: This table reports estimates of the relationship between frontier definitions and the share of the county with any of the infectious diseases considered in [Gorodnichenko and Roland \(2016\)](#) (columns 1–2) and any illness (column 3–4). The infectious diseases of interest include tuberculosis, malaria, and typhus. The specification is otherwise similar to that in Table 1, with Census division FE and standard errors clustered using the grid-cell approach of [Bester, Conley and Hansen \(2011\)](#).

K Data Sources and Construction

Harmonization to 2010 Boundaries

We harmonize all historical Census data to the 2010 boundaries using an approach suggested in [Hornbeck \(2010\)](#). First, we intersect the county shapefiles from each of the decadal census years with the 2010 county shapefile and calculate the area of each intersection. When the 2010 county falls in one or more counties of the earlier shapefile, each piece of the 2010 county is assigned a value equal to the share of the area of the piece in the earlier county multiplied by the total value of the data for the earlier county. Then, the data for each county in 2010 is the sum of all the pieces falling within its area. This harmonization procedure would be exact if all the data from the various years are evenly distributed across county areas.

Demographic Variables and Individualism

Population density. Population/area. Digitized U.S. Census data on population for every decade in 1790–2010, from [Manson et al. \(2019\)](#). The data on area is calculated using the 2010 county shapefiles from NHGIS ([Manson et al., 2019](#)) using GIS software. The county-level population data along with other pre-2010 data are harmonized to the 2010 county boundaries and the data for intercensal years is imputed using the procedure detailed in Section A. The population density figures include slaves in the antebellum period but exclude most Native Americans throughout the frontier era as they were not counted by the Census.

Sex Ratio. Whites males/white females. The data is available for every decade in 1790–1860, 1880, and 1890. Data source: [Manson et al. \(2019\)](#).

Prime Age Adult Share. Whites aged 15–49/all whites. The data used is consistently available for every decade in 1830–1860. Data source: [Manson et al. \(2019\)](#).

Illiteracy. Illiterate whites aged above 20/whites aged over 20. The variable is available consistently for 1840 and 1850. Data source: [Manson et al. \(2019\)](#).

Immigrant Share. Foreign born/population. The variable used is available for every decade in 1820–1890 (excluding 1840). Data source: [Manson et al. \(2019\)](#).

Out of State Born Share. Out-of-state born/population. The variable is consistently available for every decade in 1850–1880. Data source: [Manson et al. \(2019\)](#).

Land inequality. Gini index using distribution of farm sizes, based on county level data on the number of farms of sizes 0–10, 10–19, 20–49, 50–99, 100–499, 500–1000, and above 1000 acres. Available for every decade in 1860–1890. Data source: [Manson et al. \(2019\)](#).

Infrequent Children Names. White Children Aged 0–10 with Non-Top-10 First Names in Division/White Children Aged 0–10 With Native-Born Parents. In addition, for the same sample, we construct additional variables by calculating the popularity of names at the county, state, and national level instead of the Census division. We use the following procedure to generate the name shares: start by restricting the sample as desired (e.g. white children aged 0–10 with native parents), then calculate the number of children in the county for each given name, then using that value identify the top 10 given names within the given administrative unit, and then accordingly count the number of children in that county with the identified top 10 names in their corresponding census division. The variable restricting to white children aged 0–10 is available for every decade in 1850–1940 (excluding 1890), with further native-parent restriction for 1850 and 1880–1940 (excluding 1890). In 1940, only 5% of individuals are asked about parental birthplace. Hence, for 95% of kids, we can only identify native-born parents if those

parents co-reside with the children. For the remaining 5% of kids, we can identify native-born parents even if one of the parents does not reside with the children.

To give some examples of names, in 1850 the top 10 boy names nationally in descending order of popularity were John, William, James, George, Charles, Henry, Thomas, Joseph, Samuel and David. Meanwhile, a random sample of less common names (outside the top 25) includes ones like Alfred, Nathan, Patrick, Reuben, Herbert, Matthew, Thaddeus and Luke. For girls, the top 10 include Mary, Sarah, Elizabeth, Martha, Margaret, Nancy, Ann, Susan, Jane, and Catherine while less common names (outside the top 25) include ones like Rachel, Susannah, Nina, Olive, Charlotte, Lucinda, and Roxanna. By 1880, the rankings shifted only slightly for boys with Samuel falling outside the top 10 and Harry entering. For girls, the changes were a bit more dramatic with the new top 10 list being Mary, Sarah, Emma, Ida, Minnie, Anna, Annie, Martha, Cora, and Alice. Data source: US 100% samples (Ruggles et al., 2019) of the North Atlantic Population Project (NAPP) from Minnesota Population Center (2019) for 1850 and 1880, and complete-count restricted-access data from IPUMS (Ruggles et al., 2019) for other years.

Non-Patronymic/-Matronymic Names. Using the same sample definitions as with infrequent names, we compute the share of kids with names that are distinct from their parents' names of the given gender. This measure can only be defined for children with co-resident parents of the given gender.

Economic Status. We measure economic status using the occupational score (*occscore*) provided by IPUMS (Ruggles et al., 2019). Both measures range from 0 to 100, and capture the income returns associated with specific occupations in the 1950 Census while the *sei* measure additionally captures notions of prestige as well as educational attainment. Data source: Ruggles et al. (2019).

Public Goods, Local Taxation and Spending

Local Tax Revenue Per Capita. County taxes and other local taxes (town, city, etc) divided by county population. Data Source: Manson et al. (2019).

Local Public Debt Per Capita. County-level public debt and other local debt (town, city, etc) divided by county population. Data Source: Manson et al. (2019).

Infrastructure. Rail Access and Canal Access are indicator variables for whether the county is intersected by a railroad line or a canal, respectively. Data source: Atack et al. (2010).

Access to Irrigation Dam is an indicator on whether the county has an irrigation dam within its boundaries. We also generate similar variables separately by ownership of the dams (federal, state, local and private). Data source: Washington (2018).

Survey-Based Cultural Outcomes

Some of our key measures of contemporary preferences for government policy are based on data from multiple rounds of three widely used, nationally representative surveys: the Cooperative Congressional Election Study (CCES), the General Social Survey (GSS), and the American National Election Study (ANES). These surveys are staples in the social science literature on political preferences and social norms. For instance, Acharya, Blackwell and Sen (2016) uses CCES and ANES in a related methodological setting, and Alesina and Giuliano (2011) conducts a thorough investigation of the determinants of preferences for redistribution using the GSS. The CCES is a web-based survey conducted every two years, the ANES is an in-person survey conducted annually since 1948, and the GSS is an in-person survey conducted annually since 1972. All three are repeated cross-sections.

One advantage of working with three surveys is that we can cross-validate the findings across surveys that ask different questions about similar underlying preferences. For example, the CCES asks respondents if and how respondents would like state-level welfare spending to change whereas the ANES

asks respondents if and how federal spending on the poor should change. The CCES also includes a set of questions on policy issues such as gun ownership that are particularly relevant to some of the mechanisms driving the persistence of frontier culture. For all measures, we link county-level identifiers in the underlying data to the 2010 county boundaries.

Despite their rich level of detail, these surveys have one important limitation for our purposes, namely the limited geographic scope. The three surveys are nationally representative, but their coverage differs. While the CCES has broad spatial coverage, the GSS and ANES do not (see Appendix Figures K.1). Despite its broader coverage, the CCES has the potential disadvantage that it captures an internet-savvy sample that may not be reflective of the underlying population in the way that an in-person survey generally would. This is particularly disadvantageous given our focus on county-level variation in TFE across a swathe of the United States outside of major coastal population centers.

Prefers Cutting Public Spending On Poor. The Prefers Cutting Public Spending On Poor is an indicator variable based on the following survey question: *"Should federal spending be increased, decreased, or kept about the same on poor people?"* The variable takes a value of 1 if the respondent answered "decreased" and 0 otherwise, and it is available for 1992 and 1996. Data source: [The American National Election Studies Cumulative Data \(2012\)](#). The ANES is a large, nationally-representative survey of the American electorate in the United States taken during the presidential and midterm election years. See Appendix Figure K.1(a) for the map of the maximum survey coverage in the final sample of ANES data merged with the frontier related data.

Prefers Cut Public Spending on Welfare. This is an indicator variable based on the following survey question: *"State legislatures must make choices when making spending decisions on important state programs. Would you like your legislature to increase or decrease spending on Welfare? 1. Greatly Increase 2. Slightly Increase 3. Maintain 4. Slightly Decrease 5. Greatly Decrease."* Prefers Cut Public Spending on Welfare takes a value of 1 if the respondent answered "Slightly Decrease" or "Greatly Decrease" and 0 otherwise. The data is available in the 2014 and 2016 waves. Data source: Cooperative Congressional Election Study ([Ansolabehere and Schaffner, 2017](#)) Common Content surveys. The CCES was formed in 2006, through the cooperation of several academic institutions, to study how congressional elections, representation and voters' behavior and experiences vary with political geography and social context using very large scale national surveys. The 2014 and 2016 CCES surveys were conducted over the Internet by YouGov using a matched random sample methodology. The Common Content portion of the survey, which contains our variables of interest, surveyed 56,200 adults in 2014 and 64,600 adults in 2016. See Appendix Figure K.1(b) for the map of the maximum survey coverage in the final sample of GSS data merged with frontier-related data.

Believes Government Should Redistribute. Based on the following survey question: *"Some people think that the government in Washington ought to reduce the income differences between the rich and the poor, perhaps by raising the taxes of wealthy families or by giving income assistance to the poor. Others think that the government should not concern itself with reducing this income difference between the rich and the poor. Here is a card with a scale from 1 to 7."* We have recoded the variable so that it is increasing in preference for redistribution, where a score of 1 means that the government should not concern itself with reducing income differences and a score of 7 means the government ought to reduce the income differences between rich and poor. The Believes Government Should Redistribute is a normalized version of the above variable, and it is available in our sample for 1993 and all even years between 1994-2016. Data source: The General Social Survey ([Smith, Marsden, Hout and Kim, 2015](#)). The GSS is a repeated cross-sectional survey of a nationally representative sample of non-institutionalized adults who speak either English or Spanish. The surveys has been conducted since 1972, almost every year between 1972-1993 and biennial since 1994. While the sample size for the annual surveys was 1500, since 1994 the GSS administers the surveys to two samples in even-numbered years, each with a target sample size of 1500. The surveys provide detailed questionnaires on issues such as national spending priorities, intergroup relations, and confidence in institutions. See Appendix Figure K.1(c) for the map of the maximum survey coverage in the

final sample of CCES merged with frontier related data.

Prefers Reducing Debt by Cutting Spending. The variable is based on the CCES survey question: “The federal budget deficit is approximately [\$ year specific amount] this year. If the Congress were to balance the budget it would have to consider cutting defense spending, cutting domestic spending (such as Medicare and Social Security), or raising taxes to cover the deficit. Please rank the options below from what would you most prefer that Congress do to what you would least prefer they do: Cut Defense Spending; Cut Domestic Spending; Raise Taxes.”. While this question varies slightly from year to year, the underlying theme is the same. The Prefers Reducing Debt by Cutting Spending variable takes a value of 1 if the respondent chose “Cut Domestic Spending” as a first priority. The data is available for 2006-2014 (excluding 2013). Data source: [Ansolabehere and Schaffner \(2017\)](#).

Index of Preferences for Spending Cuts. The index is the principal component of nine dummy variables that take the value of 1 if the respondents answers “too much” to the following questions: “We are faced with many problems in this country, none of which can be solved easily or inexpensively. I’m going to name some of these problems, and for each one I’d like you to name some of these problems, and for each one I’d like you to tell me whether you think we’re spending too much money on it, too little money, or about the right amount. First (READ ITEM A) . . . are we spending too much, too little, or about the right amount on (ITEM)?”. The items considered are improving and protecting the environment, improving healthcare, solving big city problems, halting increasing crimes, dealing with drug addictions, improving the education system, improving conditions for blacks, military spending, foreign aid, welfare, and roads. The variable is available in our sample for 1993 and all even years between 1994-2016. Data source: [Smith et al. \(2015\)](#).

Opposes Affordable Care Act. Based on the CCES survey question: “The Affordable Health Care Act was passed into law in 2010. It does the following: Requires all Americans to obtain health insurance, Prevents insurance companies from denying coverage for pre-existing condition, Allows people to keep current health insurance and care provider, and Sets up national health insurance option for those without coverage, but allows states the option to implement their own insurance system. Would you have voted for the Affordable Care Act if you were in Congress in 2010?” The Prefers Repealing Affordable Care Act variable takes a value of 1 if the respondent answers “Yes” and 0 if the answer is “No”. The data is available for 2014. Data source: [Ansolabehere and Schaffner \(2017\)](#).

Opposes Increasing Minimum Wage. Based on the survey question: “As you may know, the federal minimum wage is currently \$5.15 an hour. Do you favor or oppose raising the minimum wage to \$7.25 an hour over the next two years, or not?”. The variable Opposes Increasing Minimum Wage takes a value of 1 if the respondent chooses “oppose” and 0 otherwise. Available in 2007. Data source: [Ansolabehere and Schaffner \(2017\)](#).

Opposes Banning Assault Rifles. Based on the CCES survey question: “On the issue of gun regulation, are you for or against for each of the following proposal? proposal: banning assault rifles”. Opposes Banning Assault Rifles takes value 1 if the respondent is against banning assault rifles and 0 otherwise. Available for 2014. Data source: [Ansolabehere and Schaffner \(2017\)](#).

Opposes EPA Regulations of CO₂ Emissions. Based on the CCES survey question “Do you support or oppose each of the following proposals? proposal: Environmental Protection Agency regulating Carbon Dioxide emissions.” The Opposes EPA Regulations of CO₂ Emissions takes one if the respondent supports the proposal and 0 the respondent opposes. Available for 2014. Data source: [Ansolabehere and Schaffner \(2017\)](#).

Cooperation vs. Self-Reliance. Based on the survey question: “I am going to ask you to choose which of two statements I read comes closer to your own opinion. You might agree to some extent with both, but we want to know which one is closer to your views: ONE, it is more important to be a cooperative person who works well with others; or TWO, it is more important to be a self-reliant person able to take care of oneself”. The Cooperation vs. Self-Reliance variable takes a value of 1 if the respondent chooses “cooperative” and 0 otherwise. Available in 1990. Data source: [The American National Election Studies](#).

Identifies As A Strong Republican. An indicator variable that takes 1 if the respondent identifies as a "Strong Republican." Available for 2007, 2012, 2014 and 2016. Data source: ([Ansolabehere and Schaffner, 2017](#)).

Other Long-run Outcomes

County Property Tax Rate. The average effective property tax rates per \$100 of value, calculated at the county level as the ratio of the average real estate tax over the average house value. Data source: The data is obtained from the National Association of Home Builders, which calculated the average effective property tax rates based on the 2010–2014 American Community Survey (ACS) data from the Census Bureau.

Republican Vote Share in Presidential Elections. Votes for a GOP candidate/total votes, at the county level. Data source: ICPSR (8611) Presidential and Congressional Races, 1840–1972 for 1900–1972 elections, General Election Data for the United States 1950–1990 (ICPSR 13) for 1976–1996 elections, and Dave Leip's Atlas of U.S. Presidential Elections for 2000–2016 elections (2017).

Geographic and Agroclimatic Controls

Land productivity measures. Average of attainable yields for alfalfa, barley, buckwheat, cane sugar, carrot, cabbage, cotton, ax, maize, oats, onion, pasture grasses, pasture legumes, potato, pulses, rice, rye, sorghum, sweet potato, tobacco, tomato, and wheat. We normalize each product's values dividing it by the maximum value for that product in the sample. Measures of attainable yields were constructed by the FAO's Global Agro-Ecological Zones project v3.0 ([IIASA/FAO, 2012](#)) using climatic data, including precipitation, temperature, wind speed, sunshine hours and relative humidity (based on which they determine thermal and moisture regimes), together with crop-specific measures of cycle length (i.e. days from sowing to harvest), thermal suitability, water requirements, and growth and development parameters (harvest index, maximum leaf area index, maximum rate of photosynthesis, etc). Combining these data, the GAEZ model determines the maximum attainable yield (measured in tons per hectare per year) for each crop in each grid cell of 0.083×0.083 degrees. We use FAO's measures of agroclimatic yields (based solely on climate, not on soil conditions) for intermediate levels of inputs/technology and rain-fed conditions.

Area. The log of surface area in square miles, calculated using the 2010 county shapefiles from NHGIS using GIS software. Data source: [Manson et al. \(2019\)](#).

Temperature. County-level mean annual temperature measured in Celsius degrees. Data source: [IIASA/FAO \(2012\)](#).

Rainfall. County-level average annual precipitation measured in mm. Data source: [IIASA/FAO \(2012\)](#).

Elevation. County-level average terrain elevation in km. Data source: [IIASA/FAO \(2012\)](#).

Latitude. Absolute latitudinal distance from the equator in decimal degrees, calculated from the centroid of each county using GIS software and county shapefiles from NHGIS. Data source: [Manson et al. \(2019\)](#).

Longitude. Absolute longitudinal distance from the Greenwich Meridian in decimal degrees, calculated from the centroid of each county using GIS software and county shapefiles from NHGIS. Data source: [Manson et al. \(2019\)](#).

Distance to the coastline, rivers, and lakes. Minimum distance to a point in the coastline, rivers, and lakes in km, calculated from the centroid of each county using GIS software and county shapefiles from NHGIS. Data source: [Manson et al. \(2019\)](#).

Additional Variables

Annual Migration Inflow. Total number of migrants entering the United States every year. The data for 1820–1890 is available from the [Migration Policy Institute \(2016\)](#), which tabulates data from the Office of Immigration Statistics, while the data for 1790–1819 is imputed from [Tucker \(1843\)](#). To construct the instrumental variable based on annual migration inflows predicted by weather shocks in Europe, we use the annual migration inflows to the U.S. from Belgium, Denmark, England, France, Germany, Greece, Ireland, Italy, Norway, Poland, Portugal, Russia, Scotland, Spain, Sweden, Wales from 1820–1890. Data source: [Willcox \(1929\)](#).

Years Connected to Railroad by 1890. The number of years since the county is first intersected by railroad to 1890. Data source: [Atack et al. \(2010\)](#).

Birthplace diversity, 1890. We take $1 - \sum_o (\text{birthplace}_{oc} / \text{population}_c)^2$, which is simply 1 minus the Herfindahl concentration index for origin o birthplace diversity in county c in 1890. Birthplaces include US or a given country or country grouping abroad. Data source: [Manson et al. \(2019\)](#)

Ruggedness. County-level average Terrain Ruggedness Index computed using 30-arc grid data on terrain variability. Data source: [Nunn and Puga \(2012\)](#).

Distance to nearest portage site. Minimum distance from county centroid to the nearest portage site, which is defined as the location where a river basin intersects the fall line. Data source: [Bleakley and Lin \(2012\)](#).

Manufacturing Employment Share. County-level percent of employment in manufacturing industries in 1890. Data source: [Manson et al. \(2019\)](#).

Distance to Nearest Mine. Minimum distance from county centroid to a site where there was a mineral discovery before 1890. The data is from the Mineral Resources Data System (MRDS) edited by the US Geological Survey. Data source: [McFaul et al. \(2000\)](#).

Distance to nearest Indian battle sites. Minimum distance from county centroid to major Indian battle sites. The battles sites are digitized using a map from [McFaul et al. \(2000\)](#).

Immigrant share, 1890. County-level percent of foreign born population in 1890. Data source: [Manson et al. \(2019\)](#).

Scottish and Irish immigrant share, 1890. County-level percent of population born in Scotland or Ireland in 1890. Data source: [Manson et al. \(2019\)](#)

Slave population share, 1860. County-level percent of slave population in 1860. Data source: [Manson et al. \(2019\)](#).

Rainfall Risk. Following [Ager and Ciccone \(2017\)](#), county level rainfall risk is constructed as the variance of the annual average log monthly rainfall from 1895–2000. Data source: [Oregon State University \(2018\)](#).

Linked Sample 1850–1880

This section provides a step-by-step description of how we construct the 1850 to 1880 linked sample and how we select the final sample used for the analyses in Section 5.2. The data comes from the 1850 and 1880 US 100% samples ([Ruggles et al., 2019](#)) of the North Atlantic Population Project (NAPP) from [Minnesota Population Center \(2019\)](#). The linking procedure closely follows the approach detailed in [Feigenbaum \(2016\)](#).

Constructing the linked sample

1. *Select the sample to be linked from the 1850 complete-count Census data:* First, using the complete-count 1850 Census data from NAPP, we select all white males aged 0-20 in 1850. As is common in the literature, we restrict the linking sample to only males because last names are key in the linking process and the task of linking women across censuses is much more difficult given the frequency with which women change their last names in marriage. We keep the information on each individual's first name, last name, age and birth country. Call this dataset X1.

2. *Identify all targets that could be potential matches from the 1880 census:* The linking procedure requires that for each individual in 1850, the target person in 1880 has the same gender, birth country and within +/- 3 years of the predicted age from the 1850 census. In addition, to limit the set of possible links, we impose an additional restriction that the potential link has a first name Jaro-Winkler distance of less than .2, and a last name Jaro-Winkler distance of less than .2. Call this dataset X2.

3. *Create all possible matching pairs between X1 and X2.* For each male aged 0-20 in 1850, we create a dataset that contains all the possible matching pairs with the candidates we identified in Step 2. Call this dataset X1X2. Hence, X1X2 contains all the possible match pairs for the 5,329,750 white men we extracted from the 1850 census.

4. *Create a training dataset based on manual linking of a random subset of all potential pairs:* For a random sample of 10,000 individuals in X1, we extract all the possible matches from the X1X2 dataset. The paired data contains information on first name, last name and age of the individuals both in 1850 and 1880. After discarding matches with very small link probabilities based on first and last name similarity, we are left with a random sample of potential matching pairs for 8,557 white boys from 1850 matched with all potential targets in 1880.

Then, experienced research assistants examined the randomly sampled dataset and decided which pairs formed true matches by comparing the similarities in first name, last name and age for each possible link. If there are no good matches for a given unique record in X1, then all the links were marked as non-match. The manual linking procedure generates a training dataset which contains a random subset of X1X2 in which we have determined that a record is either a match or not a match. The research assistants were able to locate 42.7% (n=3,654) of our randomly selected 8,557 white boys from 1850 in the 1880 census.

5: *Construct the training algorithm:* The algorithmic approach aims to approximate the best efforts of the researcher assistants in the manual linking procedure to assess matches. In particular, using the sample of manually linked matches from Step 4, we train an algorithm that predicts the likelihood of a match between all potential match pairs from 1850 and 1880 in the X1X2 sample.

The algorithm considers several variables that are constructed to capture similarities based on first name, last name and age similarity which aim to describe the features of the potential matches. The algorithm, which is a probit model, provides us with weights that we can put on the aforementioned variables when deciding on which links to consider as matches.

6: *Apply the algorithm generated using the training dataset to the full X1X2 data:* Using the same set of variables and the algorithm produced in Step 5, we predict the probability that each pair of matches in X1X2 is a true match. We then select which pair of matches are true matches based on an optimal cutoff that minimizes the likelihood of making false negative and false positive errors. See [Feigenbaum \(2016\)](#) for the details.

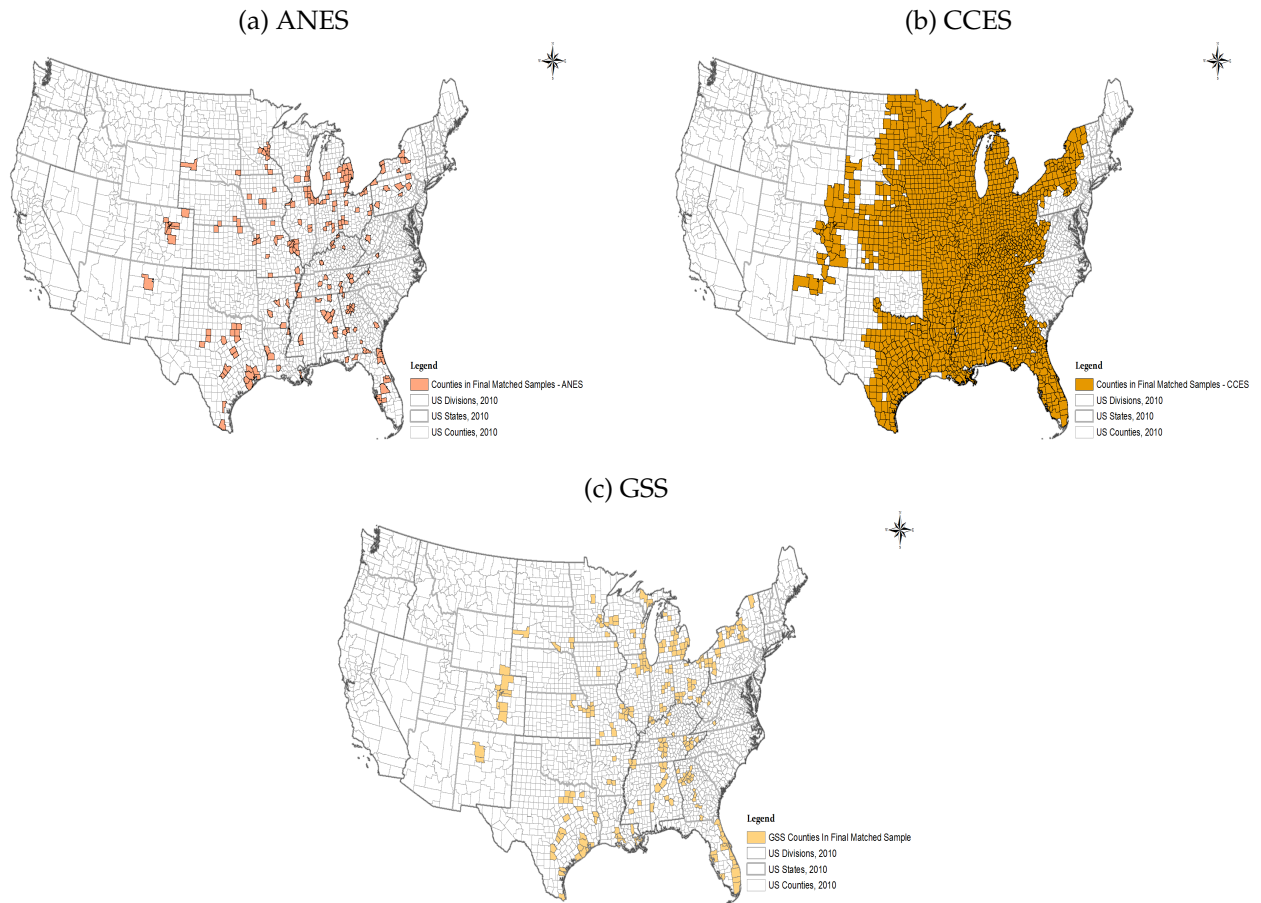
The matching process allows for multiple individuals in 1850 to be matched with a single individual in 1880. When we allow for multiple individuals from 1850 to match to a single individual in 1880, the

algorithm has a match rate of 47.9% ($n=2,552,950$). When we restrict that only one individual in 1850 can match to an individual in 1880, the algorithm has a match rate of 33.04% ($n=1,761,408$). These compare favorably to other recent efforts in the literature (see footnote 1 in Appendix I).

Selection of Sample Used in Section 5.2.

7: *Identify the individuals in the linked sample who are fathers in 1880:* We take our sample of linked white men that we traced from 1850 to 1880, and we keep those that are household heads with children in 1880. We then bring all the relevant data for our linked men both from the 1850 and 1880 census and match it to their household members in 1880. Then, we keep the data on all their children, both boys and girls, who are under the age of 20 when we observe them in 1880. Hence, the final sample used in the analysis contains the 1880 record of the children who are descendants of the men we were able to link between 1850 and 1880.

Figure K.1: Geographic Coverage For Main Survey Data Sources



Notes: Figures (a), (b), and (c) provide the geographical distribution of the maximum number of counties available in our baseline sample matched with the ANES, CCES, and GSS data, respectively. Additional counties are included when incorporating the West Coast sample or extending the historical frontier window to 1950 (see Section 4.4). Due to varying data availability across rounds, not all the counties in the above map are included in every regression using the corresponding survey data.

References

- Abramitzky, R., L. P. Boustan, and K. Eriksson**, "Europe's tired, poor, huddled masses: Self-selection and economic outcomes in the age of mass migration," *American Economic Review*, 2012, 102 (5), 1832–1856.
- Acemoglu, D., J. Moscona, and J. A. Robinson**, "State capacity and American technology: evidence from the nineteenth century," *American Economic Review: Papers & Proceedings*, 2016, 106 (5), 61–67.
- Acharya, A., M. Blackwell, and M. Sen**, "The Political Legacy of American Slavery," *The Journal of Politics*, 2016, 78 (3), 621–641.
- Ager, P. and A. Ciccone**, "Agricultural risk and the spread of religious communities," *Journal of the European Economic Association*, 2017, 16 (4), 1021–1068.
- Alesina, A. and P. Giuliano**, "Preferences for Redistribution," in J. Benhabib, A. Bisin, and M. O. Jackson, eds., *Handbook of Social Economics*, Vol. 1, North Holland, 2011, pp. 93–131.
- , **E. L. Glaeser, and B. Sacerdote**, "Why Doesn't the United States Have a European-Style Welfare State?," *Brookings Papers on Economic Activity*, 2001, 2001 (2), 187–277.
- Ansolabehere, S. and B. F. Schaffner**, "General Social Surveys, 1972–2014," *CCES Common Content*, 2006–2016, 2017.
- Atack, J., F. Bateman, M. Haines, and R. A. Margo**, "Did Railroads Induce or Follow Economic Growth?," *Social Science History*, 2010, 34 (2), 171–197.
- Bailey, M., C. Cole, and C. Massey**, "Simple Strategies for Improving Inference with Linked Data: A Case Study of the 1850–1930 IPUMS Linked Representative Historical Samples," *Historical Methods*, forthcoming.
- Bester, C. A., T. G. Conley, and C. B. Hansen**, "Inference with dependent data using cluster covariance estimators," *Journal of Econometrics*, 2011, 165 (2), 137–151.
- Billington, R. A.**, *America's Frontier Heritage*, University of New Mexico Press, 1974.
- Bleakley, H. and J. Lin**, "Portage and path dependence," *Quarterly Journal of Economics*, 2012, 127 (2), 587–644.
- Cameron, A. C., J. B. Gelbach, and D. L. Miller**, "Robust inference with multiway clustering," *Journal of Business & Economic Statistics*, 2011, 29 (2), 238–249.
- Collins, W. J. and M. H. Wanamaker**, "Up from Slavery? African American Intergenerational Economic Mobility Since 1880," *NBER Working Paper 23395*, 2017.
- Conley, T. G.**, "GMM Estimation with Cross Sectional Dependence," *Journal of Econometrics*, 1999, 92 (1), 1–45.
- Feigenbaum, J. J.**, "Automated Census Record Linking: A Machine Learning Approach," *Unpublished Manuscript*, 2016. <https://jamesfeigenbaum.github.io/research/census-link-ml/>.
- Gorodnichenko, Y. and G. Roland**, "Culture, Institutions and the Wealth of Nations," *Review of Economics and Statistics*, 2016, 99 (3), 402–416.
- Hornbeck, R.**, "Barbed wire: Property rights and agricultural development," *The Quarterly Journal of Economics*, 2010, 125 (2), 767–810.
- IIASA/FAO**, "Global Agro-ecological Zones (GAEZ v3.0)," *Unpublished Manuscript*, 2012.
- Long, J. and J. Ferrie**, "Intergenerational Occupational Mobility in Britain and the US since 1850," *American Economic Review*, 2013, 103 (4), 1109–1137.
- Manson, S., J. Schroeder, D. Van Riper, and S. Ruggles**, "IPUMS National Historical Geographic Information System: Version 14.0 [Database].," Minneapolis, MN: IPUMS. 2019.
- McFaul, EJ, GT Mason, WB Ferguson, and BR Lipin**, "US Geological Survey mineral databases; MRDS and MAS/MILS," Technical Report, US Geological Survey, 2000.
- Migration Policy Institute**, "Tabulations of U.S. Department of Homeland Security, Office of Immigration Statistics, Yearbook of Immigration Statistics," 2016.

- Minnesota Population Center**, "Integrated Public Use Microdata Series, International: Version 7.2 [dataset]," Minneapolis: IPUMS, 2019 2019.
- Nunn, N. and D. Puga**, "Ruggedness: The blessing of bad geography in Africa," *Review of Economics and Statistics*, 2012, 94 (1), 20–36.
- of Engineers: FEMA's National Dam Safety Program Washington, DC: U.S. Army Corps**, "The National Inventory of Dams," 2018.
- Oregon State University**, "PRISM Climate Group," 2018.
- Overmeyer, P. H.**, "Westward Expansion Before the Homestead Act," in H. F. Williamson, ed., *The Growth of the American Economy: An Introduction to the Economic History of the United States*, New York: Prentice-Hall, Inc, 1944, pp. 82–112.
- Robinson, P. M.**, "Root-N-Consistent Semiparametric Regression," *Econometrica*, 1988, 56 (4), 931–954.
- Ruggles, S., S. Flood, R. Goeken, J. Grover, E. Meyer, J. Pacas, and M. Sobek**, "Integrated Public Use Microdata Series: Version 9.0 [dataset]," Minneapolis, MN: IPUMS, 2019. 2019.
- Smith, T. W., P. Marsden, M. Hout, and J. Kim**, "General Social Surveys, 1972-2014," 2015.
- Stewart, J. I.**, "Migration to the Agricultural Frontier and Wealth Accumulation, 1860–1870," *Explorations in Economic History*, 2006, 43 (4), 547–577.
- The American National Election Studies**, *THE ANES GUIDE TO PUBLIC OPINION AND ELECTORAL BEHAVIOR*, University of Michigan, Center for Political Studies: Ann Arbor, MI.
- Thornhill, R. and C. L. Fincher**, *The Parasite-Stress Theory of Values and Sociality: Infectious Disease, History and Human Values Worldwide*, New York: Springer, 2014.
- Tucker, G.**, "Progress of Population and Wealth in the United States, in Fifty Years, as Exhibited by the Decennial Census Taken in that Period," *The Merchants' Magazine and Commercial Review*, 1843, 9, 136–44.
- Willcox, W. F.**, "International Migrations, Volume I: Statistics," *National Bureau of Economic Research*, 1929.